

Machine Learning-Based Analysis of Clickbait Detection in Bangla News Headlines

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ABSTRACT

The huge use of news platforms in online and social media has increased the clickbait headlines. Which is designed to attract people through misleading language and tones, and detecting such headlines is really a challenge, especially for Bangla, as this is a low-resource language because Bangla doesn't have a good amount of annotated datasets and linguistic diversity. This research presents a machine learning-based analysis of clickbait detection in Bangla news headlines. A labeled dataset that has both clickbait and non-clickbait headlines was preprocessed using the standard text-cleaning process. Here, TF-IDF was applied for feature extraction, and some machine learning classifiers, such as Naive Bayes, logistic regression, and random forest, have been trained. Model performance was achieved using accuracy, precision, recall, and F1-score. The results showed that TF-IDF-based machine learning models can identify clickbait in Bangla news headlines, which provides a good baseline for future research in Bangla clickbait detection.

Keywords

Clickbait, machine learning, TF-IDF

1. BACKGROUND AND MOTIVATION

In the era of online news and social media platforms, the rapid growth of clickbait headlines created the necessity of detecting that news. Clickbait headlines are designed to attract viewers through misleading and exaggerated language. Clickbait detection for low-resource languages like Bangla is a challenging task. The aim of this study is to explore the effectiveness of machine learning models for Bangla clickbait detection.

2. AIM AND RESEARCH QUESTION

The primary aim of this study is to show the effectiveness of machine learning models, which are traditional in detecting clickbait headlines in textual features in Bangla news. This research focuses on evaluating different machine learning models and also shows their performance to see and identify which are the suitable baseline models for Bangla clickbait detection.

2.1 Research Questions

- How effectively can machine learning models detect clickbait in Bangla news headlines?

- Which machine learning model performs best for Bangla clickbait detection using TF-IDF features?
- How do different models compare in terms of accuracy, precision, recall, and F1 score?

3. METHODOLOGY

The dataset contains Bangla news headlines and articles labeled as clickbait and non-clickbait. Preprocessing of the text data was done by tokenization, removing URLs and unnecessary symbols, and eliminating extra spaces and duplicates. TF-IDF vectorization was applied to turn the texts into numerical features, followed by training of multiple machine learning models and evaluation using metrics such as precision, recall, accuracy, and F1 score.

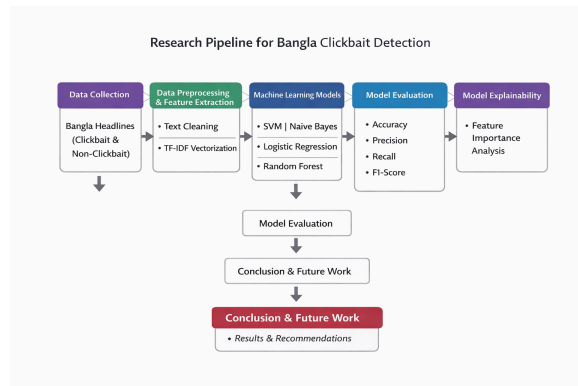


Figure 1: Pipeline of the research

3.1 Data Preprocessing and Exploratory Data Analysis

The dataset used in this study contains Bangla news headlines labeled as clickbait and non-clickbait. Initially, the headline text had noise such as punctuation, extra whitespace, and

special characters. To clean and normalize the text data, systematic preprocessing has been applied, which included converting all text to a consistent format, removing unnecessary punctuation and symbols, removing duplicate headlines, and removing extra spaces. Additionally, EDA has also been done to understand the pattern and the structure of the dataset.

The class distribution was analyzed both before and after cleaning. From Figures 2 and 3, the data imbalance can be seen, as the dataset contains more non-clickbait headlines. (0=non clickbait,1=clickbait). The headline length was also analyzed both before and after cleaning . In Figure 4 it can be seen that after preprocessing, the headline length was reduced.

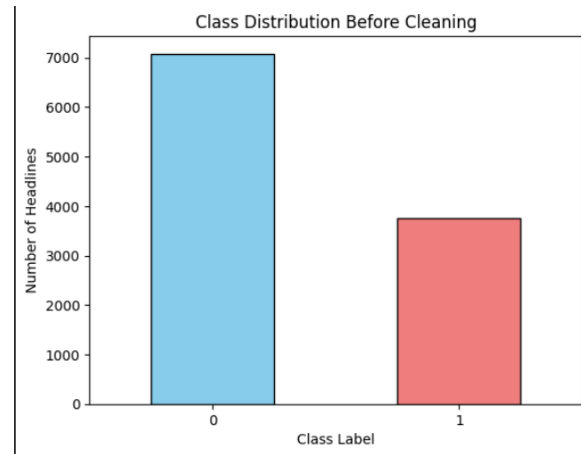


Figure 2: Class Distribution before Preprocessing

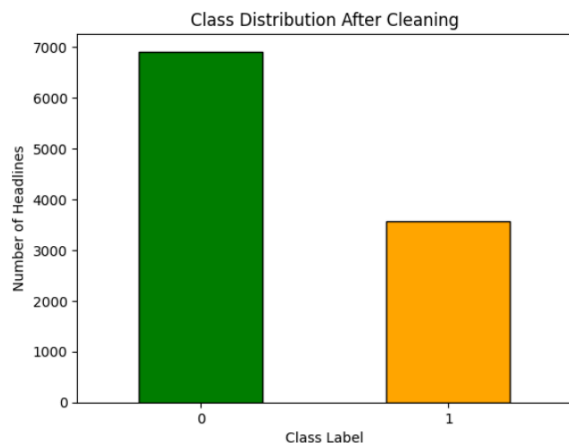


Figure 3: Class Distribution after Preprocessing

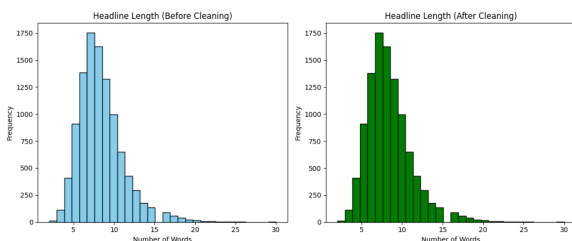


Figure 4: Headline Length Before preprocessing and after preprocessing

3.2 Machine Learning Models Used

A combination of supervised and unsupervised machine learning models was used in this study. An unsupervised model was used to explore the hidden patterns in the dataset, while supervised learning models were used for classification.

3.2.1 Logistic Regression

Logistic regression is a simple yet effective linear model that provides a baseline for binary text classification tasks.

3.2.2 Multinomial Naive Bayes

It is a commonly used probabilistic classifier used in text classification.

3.2.3 Linear Support Vector Machine (SVM)

Linear SVM is a powerful classifier that finds the optimal hyperplane that separates the two classes (clickbait or non-clickbait in this case)

3.2.4 KMeans Clustering

K-means clustering explores internal groupings within the dataset without using class labels.

4. MODEL EVALUATION

The performance of the three supervised models, along with one unsupervised model, was evaluated for Bangla clickbait detection using common metrics. While Multinomial Naive Bayes demonstrated slightly higher accuracy (70.87%) but lower recall for the clickbait class, Logistic Regression attained an accuracy of 69.21% on the validation set with balanced precision, recall, and F1-score. The accuracy of linear SVM was 67.47%, which was significantly lower. K-means clustering yielded low ARI (0.0079) and NMI (0.0006) scores, indicating poor alignment with the true labels and emphasizing the need for supervised learning in this task. Logistic regression was chosen as the final model based on interpretability and validation performance. It was tested on the test set and showed consistent generalization with an accuracy of 68.19%.

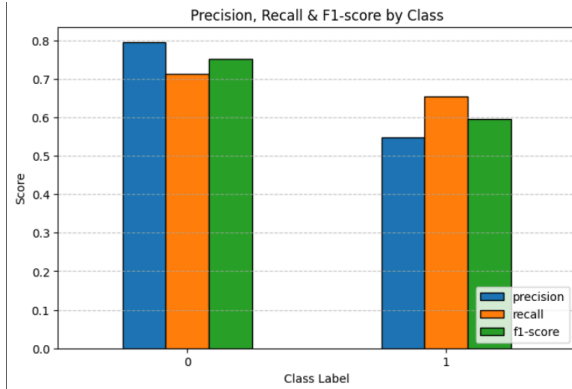


Figure 5: Performance of Logistic Regression

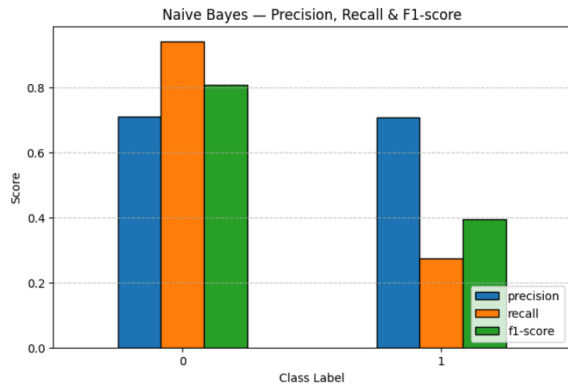


Figure 6: Performance of Naive Bayes

	Model	Accuracy (%)	Precision	Recall	F1-score
0	Logistic Regression	69.211618	0.708755	0.692116	0.697362
1	Multinomial Naive Bayes	70.871369	0.708556	0.708714	0.664510
2	Linear SVM	67.468880	0.691024	0.674689	0.680089

Figure 7: Performance of all models

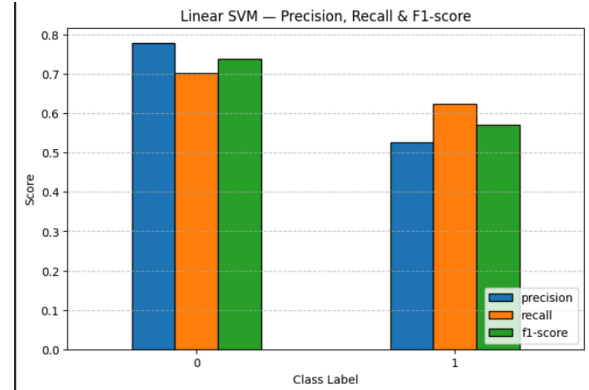


Figure 8: Performance of Linear SVM

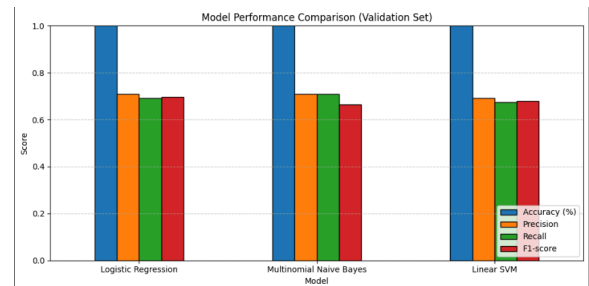


Figure 9: Performance Comparison of all models

5. MODEL EXPLAINABILITY

By examining feature coefficients obtained from TF-IDF representations, explainability was integrated utilizing the logistic regression model's built-in interpretability. The impact of individual Bengali words on clickbait prediction is shown by the model's

learned coefficients with greater positive values denoting a stronger clickbait relationship. The algorithm captures important linguistic patterns rather than random correlations. In Figure 10, frequently used sensational and attention grabbing phrases are shown.

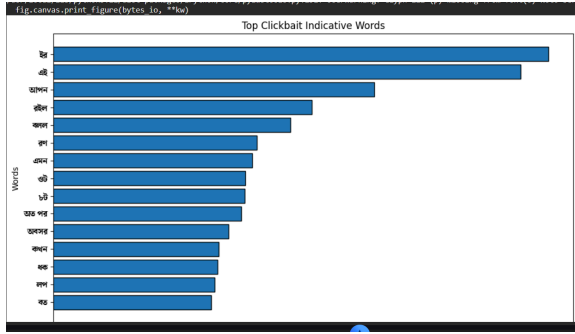


Figure 10: Top Clickbait Indicative Words

6. CONCLUSION

This study used TF-IDF features to test several machine learning techniques for Bangla clickbait detection. Among the supervised models, logistic regression achieved the most balanced performance, with an accuracy of 69.21% on the validation set and 68.19% on the test set. Although Multinomial Naive Bayes showed slightly higher accuracy, it performed poorly in detecting clickbait headlines, and Linear SVM showed lower overall results. Lastly, the study indicates that interpretable linear models are useful and practical for detecting clickbait in Bangla.

7. FUTURE WORK

Future research is to focus on improving Bangla clickbait detection by some advanced deep learning models and some Transformer-based architectures to better capture contextual information. By expanding and balancing the dataset, along with using semantic and some linguistic features, we can enhance the model's performance. However, some hybrid approaches may improve the accuracy. Additionally, developing real-time detection systems and improving model performance will make these methods more efficient and trustworthy for the online news platforms.

8. REFERENCES

- [1]Yasin Arfat and Sharmistha Chanda Tista, “Bangla Misleading Clickbait Detection Using Ensemble Learning Approach,” pp. 184–189, May 2024, doi: <https://doi.org/10.1109/iceeict62016.2024.10534333>.

Colab	file	link-
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